

COMSATS UNIVERSITY ISLAMABAD

ATTOCK CAMPUS



DEPARTMENT OF COMPUTER SCIENCE

COURSE	SQE Lab
INSTRUCTOR	Ms. Munazza
TASK	Assignment 4

Submitted By:

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Academic Task Tracker System

Introduction:

Academic Task Tracker System is a Python-based software application developed to help students manage their academic tasks. This system allows users to manage assignments, quizzes, mids, terminals, and project activities. Users can add tasks, view tasks, search tasks, update task status, and delete tasks. The system helps students stay organized and track their academic progress

Tools Used:

Python – Used for system development.

Visual Studio Code – Used for writing and running code.

SonarQube – Used for code quality analysis.

Snyk – Used for security analysis.

[GitHub](#) – Used for version control and project hosting.

Core Features:

1. Add Task
2. View Tasks
3. Search Task
4. Update Task Status
5. Delete Task
6. Exit System

Manual Test Cases:

Test Case ID	Feature	Input	Expected Output	Actual Output	Status
TC-01	Add Task	PF, Assignment, Pending	Task added successfully	Task added successfully	Pass
TC-02	View Tasks	Press 2	Tasks displayed	Tasks displayed	Pass
TC-03	Search Task	PF	Task Found	Task Found	Pass

TC-04	Update Status	PF, Completed	Status updated	Status updated	Pass
TC-05	Delete Task	PF	Task deleted	Task deleted	Pass
TC-06	Exit	Press 6	Program closed	Program closed	Pass

Bug Report :

Bug ID	Feature	Problem Found	Expected Behavior	Actual Behavior	Status
BUG-01	View Tasks	After restarting program, tasks disappear	Tasks should remain available	Tasks list becomes empty	Fixed
BUG-02	Search Task	Wrong course name entered	Show not found message	Task not found shown	Fixed

Program execution screenshots:

```

PS C:\Users\JHP\OneDrive\Desktop\Academic_Task_Tracker> git push origin main --force
PS C:\Users\JHP\OneDrive\Desktop\Academic_Task_Tracker> python main.py

===== Academic Task Tracker =====
1. Add Task
2. View Tasks
3. Search Task
4. Update Task Status
5. Delete Task
6. Exit
Enter your choice: 1
Enter course name: Sqa Lab
Enter task type (Assignment/Quiz/Mid/Terminal/Project): Project
Enter status (Pending/Ongoing/Completed): Completed
Task added successfully!

===== Academic Task Tracker =====
1. Add Task
2. View Tasks
3. Search Task
4. Update Task Status
5. Delete Task
6. Exit
Enter your choice: 2

===== All Tasks =====
Course: Sqa Lab
Task Type: Project
Status: Completed
=====

===== Academic Task Tracker =====
1. Add Task
2. View Tasks
3. Search Task
4. Update Task Status
  
```

```
File Edit Selection View Go Run ... Academic_Task_Tracker Update
EXPLORER PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
ACADEMIC_TASK_TRACKER
> .sonar
main.py
requirements.txt
PS C:\Users\HP\OneDrive\Desktop\Academic_Task_Tracker> python main.py

===== Academic Task Tracker =====
1. Add Task
2. View Tasks
3. Search Task
4. Update Task Status
5. Delete Task
6. Exit
Enter your choice: 3
Enter course name to search: Sqe Lab

Task Found!
Course: Sqe Lab
Task Type: Project
Status: Completed

===== Academic Task Tracker =====
1. Add Task
2. View Tasks
3. Search Task
4. Update Task Status
5. Delete Task
6. Exit
Enter your choice: 4
Enter course name to update: SQE LAB
Enter new status: PROJECT IS COMPLETED
Status updated successfully!

===== Academic Task Tracker =====
1. Add Task
2. View Tasks
3. Search Task
4. Update Task Status
5. Delete Task
6. Exit
```

```
File Edit Selection View Go Run ... Academic_Task_Tracker Update
EXPLORER PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
ACADEMIC_TASK_TRACKER
> .sonar
main.py
requirements.txt
PS C:\Users\HP\OneDrive\Desktop\Academic_Task_Tracker> python main.py
Status: Completed

===== Academic Task Tracker =====
1. Add Task
2. View Tasks
3. Search Task
4. Update Task Status
5. Delete Task
6. Exit
Enter your choice: 4
Enter course name to update: SQE LAB
Enter new status: PROJECT IS COMPLETED
Status updated successfully!

===== Academic Task Tracker =====
1. Add Task
2. View Tasks
3. Search Task
4. Update Task Status
5. Delete Task
6. Exit
Enter your choice: 5
Enter course name to delete: SQE LAB
Task deleted successfully!

===== Academic Task Tracker =====
1. Add Task
2. View Tasks
3. Search Task
4. Update Task Status
5. Delete Task
6. Exit
Enter your choice: 6
Program closed.
PS C:\Users\HP\OneDrive\Desktop\Academic_Task_Tracker>
```

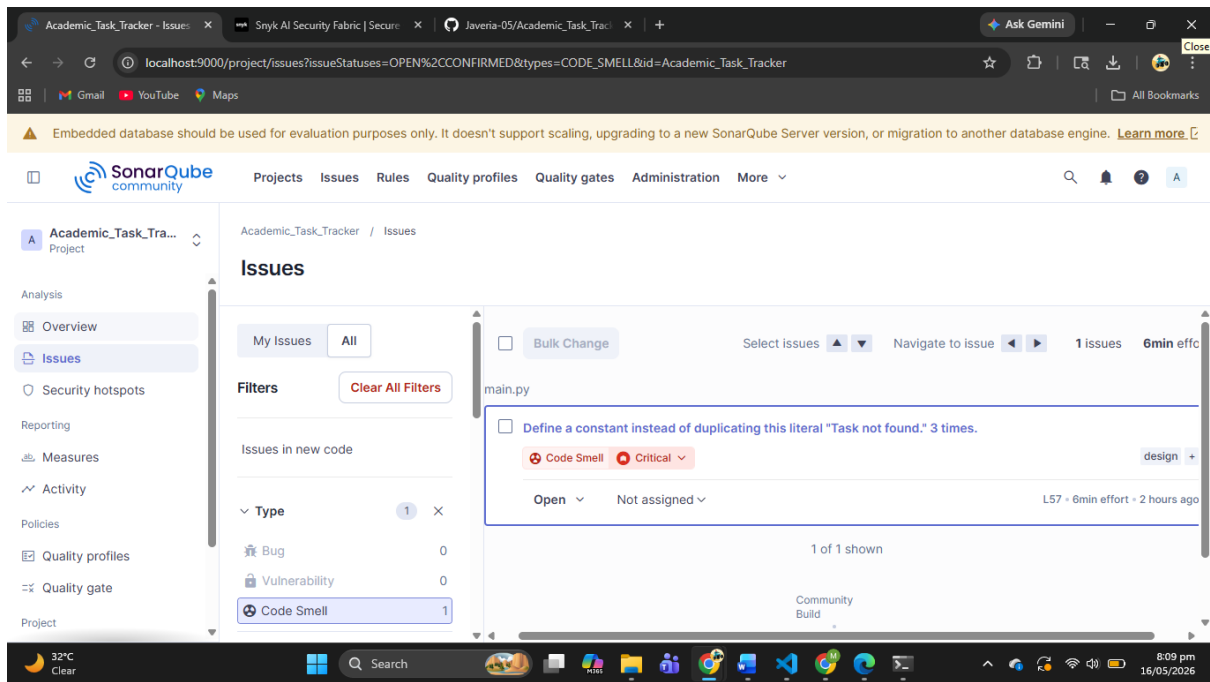
SonarQube analysis screenshots:

This screenshot shows the SonarQube Overview page for the project 'Academic_Task_Tracker'. The page indicates that the quality gate has been passed, with a green checkmark and the word 'Passed' in large text. A warning message states: 'The last analysis has warnings. See details'. The overview shows 82 Lines of Code, Version not provided, and Last analysis 2 hours ago. The left sidebar lists various analysis categories: Overview (selected), Issues, Security hotspots, Reporting, Measures, Activity, Policies, Quality profiles, and Quality gate. The bottom section displays three metrics: Vulnerabilities (0 Open issues), Bugs (0 Open issues), and Code Smells (1 Open issues). The Windows taskbar at the bottom shows the time as 7:58 pm on 16/05/2026.

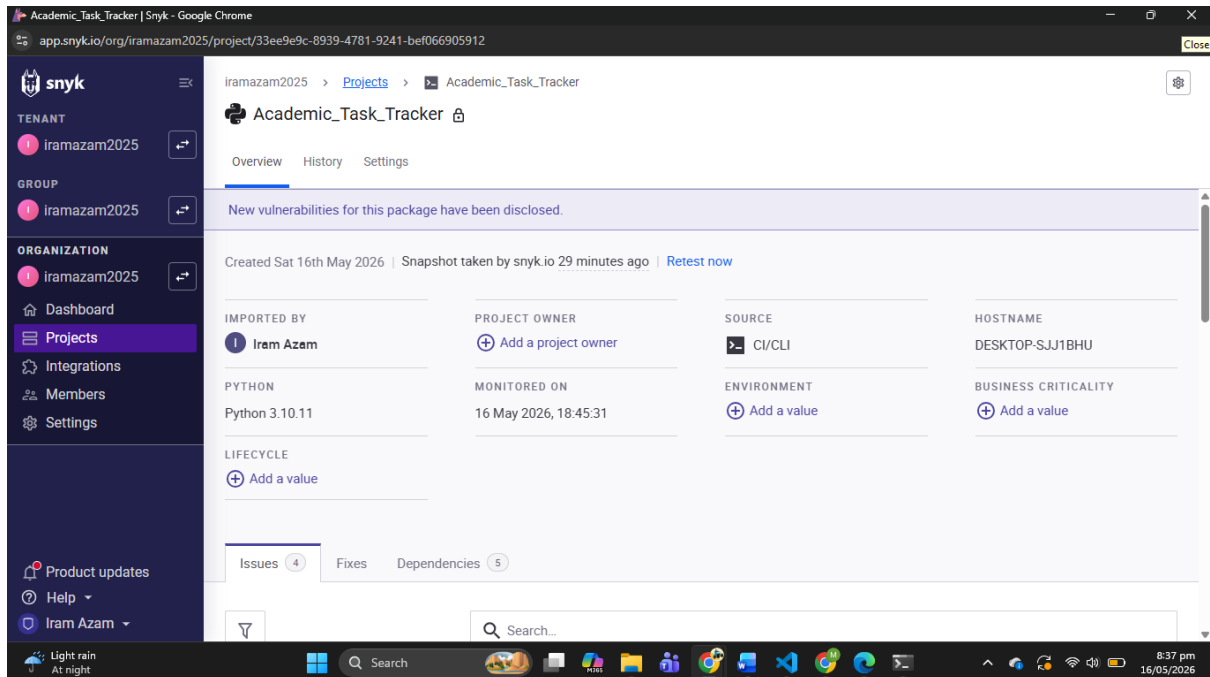
Metric	Status
Vulnerabilities	0 Open issues
Bugs	0 Open issues
Code Smells	1 Open issues

This screenshot shows the SonarQube Overview page for the project 'Academic_Task_Tracker' with more detailed metrics. The quality gate status is still 'Passed'. The overview shows 82 Lines of Code, Version not provided, and Last analysis 2 hours ago. The left sidebar is the same as the first screenshot. The bottom section displays six metrics: Vulnerabilities (0 Open issues), Bugs (0 Open issues), Code Smells (1 Open issues), Accepted issues (0), Coverage (0.0%), and Duplications (0.0%). The Windows taskbar at the bottom shows the time as 8:09 pm on 16/05/2026.

Metric	Status
Vulnerabilities	0 Open issues
Bugs	0 Open issues
Code Smells	1 Open issues
Accepted issues	0
Coverage	0.0%
Duplications	0.0%



Snyk security scan screenshots:



Academic_Task_Tracker | Snyk - Google Chrome
app.snyk.io/org/iramazam2025/project/33ee9e9c-8939-4781-9241-bef066905912

snyk
TENANT
iramazam2025
GROUP
iramazam2025
ORGANIZATION
iramazam2025
Dashboard
Projects
Integrations
Members
Settings
Product updates
Help
Iram Azam

iramazam2025 > Projects > Academic_Task_Tracker

Academic_Task_Tracker

Overview History Settings

Issues (4) Fixes Dependencies (5)

Search...

4 of 4 issues Group by none Sort by highest priority score

H **urllib3** - Decompression Bomb [🔗](#) SCORE **731**
VULNERABILITY | ...

Introduced through requests@2.32.3 Exploit maturity NO KNOWN EXPLOIT
Fixed in urllib3@2.7.0

Show more detail

Ignore

SEVERITY
☐ Critical 0
☐ High 2
☐ Medium 2
☐ Low 0

PRIORITY SCORE
Scored between 0 - 1000

"FIXED IN" AVAILABLE
☐ Yes 4

Light rain At night 8:37 pm 16/05/2026

Academic_Task_Tracker | Snyk - Google Chrome
app.snyk.io/org/iramazam2025/project/33ee9e9c-8939-4781-9241-bef066905912

snyk
TENANT
iramazam2025
GROUP
iramazam2025
ORGANIZATION
iramazam2025
Dashboard
Projects
Integrations
Members
Settings
Product updates
Help
Iram Azam

iramazam2025 > Projects > Academic_Task_Tracker

Academic_Task_Tracker

Overview History Settings

☐ No 0

COMPUTED FIXABILITY
☐ Fixable 4
☐ Partially fixable 0
☐ No supported fix 0

EXPLOIT MATURITY
☐ Mature 0
☐ Proof of concept 1
☐ No known exploit 3
☐ No data 0

CVSS 4 EXPLOIT MATURITY
☐ Attacked 0

H **urllib3** - Insertion of Sensitive Information Into Sent Data [🔗](#) SCORE **696**
VULNERABILITY | ...

Introduced through requests@2.32.3 Exploit maturity NO KNOWN EXPLOIT
Fixed in urllib3@2.7.0

Show more detail

Ignore

M **requests** - Insertion of Sensitive Information Into Sent Data [🔗](#) SCORE **606**
VULNERABILITY | ...

Introduced through requests@2.32.3 Exploit maturity PROOF-OF-CONCEPT
Fixed in requests@2.32.4

Show more detail

Rain coming 9:12 PM 8:37 pm 16/05/2026

Academic_Task_Tracker | Snyk - Google Chrome
app.snyk.io/org/iramazam2025/project/33ee9e9c-8939-4781-9241-bef066905912

snyk
TENANT
iramazam2025
GROUP
iramazam2025
ORGANIZATION
iramazam2025
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Iram Azam

iramazam2025 > Projects > Academic_Task_Tracker

Academic_Task_Tracker

Overview History Settings

☐ Attacked 0
☐ Proof of concept 0
☐ Not defined 0

STATUS
☒ Open 4
☐ Patched 0
☐ Ignored 0

Fixed in requests@2.32.4
Show more detail

Ignore

M requests - Insecure Temporary File
VULNERABILITY | ...
SCORE 419

Introduced through requests@2.32.3
Fixed in requests@2.33.0
Show more detail
Learn about this type of vulnerability

Exploit maturity NO KNOWN EXPLOIT

Ignore

Rain coming 9:12 PM
Search
8:38 pm 16/05/2026

Academic_Task_Tracker | Snyk - Google Chrome
app.snyk.io/org/iramazam2025/project/33ee9e9c-8939-4781-9241-bef066905912

snyk
TENANT
iramazam2025
GROUP
iramazam2025
ORGANIZATION
iramazam2025
Dashboard
Projects
Integrations
Members
Settings
Product updates
Help
Iram Azam

iramazam2025 > Projects > Academic_Task_Tracker

Academic_Task_Tracker

Overview History Settings

Issues (4) Fixes Dependencies (5)

Fix Advice

Upgradable issues (2)

> Pin transitive dependency urlib3@2.6.3 to version urlib3@2.7.0

> Upgrade direct dependency requests@2.32.3 to version requests@2.33.0

Rain coming 9:12 PM
Search
8:38 pm 16/05/2026

Academic_Task_Tracker | Snyk - Google Chrome

app.snyk.io/org/iramazam2025/project/33ee9e9c-8939-4781-9241-bef066905912

iramazam2025 > Projects > Academic_Task_Tracker

Academic_Task_Tracker

Overview History Settings

Issues 4 Fixes Dependencies 5

Search...

DEPENDENCY	LATEST	LAST PUBLISHED	ISSUES	LICENSES	VULNERABLE PATHS
certifi@2026.2.25			0 C 0 H 0 M 0 L		0
charset-normalizer@3.4.7			0 C 0 H 0 M 0 L		0
idna@3.11			0 C 0 H 0 M 0 L		0
requests@2.32.3			0 C 0 H 2 M 0 L		1
urllib3@2.6.3			0 C 2 H 0 M 0 L		1

<< < 1 > >>

Rain coming 9:12 PM

8:39 pm 16/05/2026

GitHub repository screenshots:

Academic_Task_Tracker - Issues x Javeria-05/Academic_Task_Tracker x

github.com/Javeria-05/Academic_Task_Tracker/tree/main

Javeria-05 / Academic_Task_Tracker

Code Issues Pull requests Agents Actions Projects Wiki Security and quality Insights Settings

Academic_Task_Tracker (Public)

main 1 Branch 0 Tags

Go to file Add file Code

Javeria-05 Create README.md c8b2227 · now 2 Commits

.sonar	Initial project upload	1 hour ago
README.md	Create README.md	now
main.py	Initial project upload	1 hour ago
requirements.txt	Initial project upload	1 hour ago

README

Academic Task Tracker

About

No description, website, or topics provided.

Readme Activity 0 stars 0 watching 0 forks

Releases

No releases published Create a new release

Packages

30°C Clear

8:45 pm 16/05/2026

